
Optimization Methods - Gradient Descent

Do the following:

1. What are a scoring function and a loss function? Explain each with a suitable example.
2. Differentiate between simple loss, hinge loss, and cross-entropy loss.
3. **Gradient Descent**
 - (a) Explain the different types of Gradient Descent.
 - (b) Implement **Batch Gradient Descent (Vanilla Gradient Descent)** in Python using an example different from the examples given in the lecture.
 - (c) What is the **bias trick**? Explain its purpose and implement Gradient Descent using the bias trick. Show the results.
 - (d) What is **Mini-Batch Gradient Descent**? Show how it differs from Batch Gradient Descent and implement it in Python. ““
4. What are the main steps involved in the **training of a Neural Network**? Explain each step briefly.
5. What is the role of the learning rate α in updating the parameters W and $\vec{b}$? Explain the effect of using a learning rate that is too small or too large.

Note: Submit the assignment with printed screenshot captured for each question, Also print this assignment and attach with your assignment.